3 **(a)** Electric potential at a point is defined as the work done per unit positive charge by an external agent in bringing a charged body from infinity to that point. B1

(b) **(i)** The charges in the 2 spheres will re-distribute themselves between the spheres. B1

Both spheres eventually reach the same potential. B1

(ii) $q_1 + q_2 = q_1' + q_2'$ M1

$V_1\, 4\pi\varepsilon_o\, r_1 + V_2\, 4\pi\varepsilon_o\, r_2 = V\, 4\pi\varepsilon_o\, (r_1 + r_2)$

$V = (V_1\, r_1 + V_2\, r_2) / (r_1 + r_2)$ A1

Assumption:

The spheres are treated as point charges.

Or no leakage of charge to the surrounding B1

(c) **(i)** $q^2 / 4\pi\varepsilon_o\, (\mathrm{L}\sin\theta)^2 + [q^2 / 4\pi\varepsilon_o\, (2\mathrm{L}\sin\theta)^2]$ M1

$= 5q^2 / (16\pi\varepsilon_o \mathrm{L}^2 \sin^2\theta)$

$= 1.12 \times 10^{10}\, q^2 / (\mathrm{L}\sin\theta)^2$ A1

(ii) $T \sin\theta = 5q^2 / 16\pi\varepsilon_o\, \mathrm{L}^2 \sin^2\theta$ M1

$T = 5q^2 / 16\pi\varepsilon_o \mathrm{L}^2 \sin^3\theta$

$= 1.12 \times 10^{10}\, q^2 / (\mathrm{L}^2 \sin^3\theta)$ A1

(iii) Spheres A and C are deflected <u>equally</u> to higher positions as the angle θ is increased by the same amount for A and C as the force is increased. B1

Sphere B remains stationary. B1

DO NOT WRITE IN THIS MARGIN

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4 (a)